

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Partnered with Product, Data Science, and business stakeholders to translate ambiguous business questions into agent-ready specs, then designed and shipped to production a self-service analytics agent over 67 BigQuery tables (sub-agent orchestration, 8 deterministic SQL rules, 23-case golden-query evaluation suite, human-in-the-loop validation). Approved by Sr. Director and product leadership; currently the only AI skill in active use by business teams on Data Ventures.
- Designed, prototyped, and operationalized an autonomous Jira ticket-resolution agent (generative AI + agentic RAG, FAISS + BM25 retrieval over 11,000 historical SQL queries, 6-gate triage, self-healing execution); closed 400+ data requests across 8 categories over 15 months and cut routine turnaround from 30–60 minutes to under 10 minutes per request — a measurable, sustained step-change in operational performance.
- Built a registry-governed semantic layer over BigQuery with deterministic validation, value canonicalization, and a human review gate; designed for sensitive, governed data so the same business question returns the same answer every time.
- Automated ~20 recurring recruitment workflows on a 68-table BigQuery platform; stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates and reduced a recurring 3-hour task to ~3 minutes of setup.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- Delivered a RAG-to-proposal generator for an enterprise consulting client, deployed on AWS EC2; cut proposal-development time ~30%; **Best in Show — UChicago MSADS Capstone**. Worked directly with the client to scope the problem and the analytical approach end-to-end.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; designed controlled A/B tests that persuaded 15% to donate to an opposing cause.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); structured the problem with district stakeholders and drove retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit front-end + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Consulting & analysis: problem structuring, hypothesis-driven analysis, quantitative & financial analysis, stakeholder management, client-ready synthesis & presentations, end-to-end delivery and implementation

AI / analytics: machine learning, generative AI, RAG (embeddings + FAISS + BM25 + RRF), agent / sub-agent orchestration, evaluation suites (golden sets, human-in-the-loop), MCP server design, multi-model routing

Data / SQL: BigQuery, dbt, SQL (complex query design + optimization), Python, PySpark, Airflow, Databricks, MySQL

LLMs: Claude (Opus / Sonnet / Haiku), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

Viz: Power BI (DAX, XMLA TMSL), Tableau, LookML, Streamlit, Matplotlib

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL